

Ernesto Rivera

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EDUCATION

Boston University

Boston, MA

B.A. in Computer Science, Minor in Data Science

May 2026

- Relevant Coursework: Data Structures, Algorithms, Machine Learning, Database Systems, Software Engineering, Artificial Intelligence, Computational Probability, Logic Design, Operating Systems

EXPERIENCE

Whiting-Turner Contracting Co.

Irvine, CA

Data Architecture Engineer

May 2024 - Oct. 2024

- Developing a scalable task management application by architecting the backend and optimizing system logic, reducing scheduling delays by **12%**, while creating a robust technical framework for junior interns to expand.
- Restructured a **1 billion+** data point database into job site-specific databases, using **Azure IoT Hub**, **Stream Analytics**, **Cosmos DB**, and **Blob Storage**, resulting in **80%** faster query times and enhanced scalability.
- Designed and implemented a scalable email automation system using **Power Automate**, decreasing the quality control management by **90x** and optimizing project management for a large-scale research facility.

AI Software Engineer

May 2023 - Jun. 2023

- Developed a full-stack mobile resume scanner with Mindee's **OCR**, **FastAPI**, and **React Native**, significantly reducing transcription time by **25x**, employing **LLM** to accurately convert OCR data into structured **JSON**.
- Led AI adoption by creating MVPs and proof-of-concept documentation, pitching to senior management, while collaboratively developing a **Waterfall methodology** framework with guiding documentation.

Global Scope Research Lab

Boston, MA

Research Assistant

Jan. 2024 - May 2024

- Analyzed a **32 GB**, 370 million-row dataset from the US Patent Directory to uncover innovation trends before and after Hurricane Katrina, shedding light on disaster-driven innovation dynamics.
- Collaborated with a Research Assistant using **NumPy** and **Dask** on a shared computing cluster, streamlining data processing to extract scalable insights into patent trends and support post-disaster policy analysis.

MyTA

Remote

Startup Founding Engineer

Jul. 2023 - Mar. 2024

- Engineered an educational chatbot by integrating **Meta AI's Nougat** for high-accuracy OCR, and leveraging **Pinecone's** vectorization capabilities for efficient and accurate data handling.
- Refined chatbot functionality to process entire professorial materials via **OCR-to-MultiMarkdown** translation, enabling dynamic, personalized student support through **vectorized** content integration.
- Collaborated in a team of 4, liaising with **9 universities**, to catalyze interest in adopting our application.

PROJECTS AND COMPETITIONS

OCR Digit Machine Learning Classifier

Jun. 2024

- Developed an OCR application using a **custom K-Nearest Neighbors (KNN)** classifier to identify digits from **60,000 MNIST** images, achieving **95%** accuracy by optimizing preprocessing and feature extraction methods.
- Enhanced the application with a user-friendly **GUI** built using **Tkinter** and **PIL** in **Python**, utilizing ImageOps, ImageFilter, and Numpy for real-time digit recognition to demonstrate machine learning capabilities.

Gloo Hackathon

Boulder, CO

Best Technology \$50,000 Prize

Oct. 2023

- Pioneered the development of personalized faith-based AI counselors, accessible via targeted social media and search engine ads, which significantly enhanced outreach efficiency and achieved a **100x** cost reduction.
- Engineered an AI-driven **MetaHuman** backend with **Inworld AI** and **Unreal Engine**, enabling immersive **3D** interaction that improved user engagement, leveraging human-to-LLM retention stats.

Cornell Health Hackathon

New York, NY

Finalist

Feb. 2023

- Constructed an **AI-powered VR/AR** therapy app that replicates individualized exposure therapy scenarios.
- Developed and integrated a **JavaScript**-based back-end to leverage **GPT-3's API**, generating personalized prompts for **Stable Diffusion** to create immersive 360° monoscopic images.

SKILLS

Languages & Web Tech: Python, JavaScript, Java, C++, JSON, HTML, CSS, Assembly, OCaml, C, $\text{\LaTeX}$

Developer Tools: Git, GitLab, React, Firebase, OCR (Nougat, MMOCR, DocTR), FastAPI, Bash, Docker, Ngrok, Google Cloud Platform, Pandas, NumPy, Autodesk Fusion 360